

LAB # 08: Camera Calibration in Computer Vision

Lab Objective:

To estimate the intrinsic (focal length, principal point, distortion) and extrinsic (rotation, translation) parameters of a camera using a checkerboard pattern.

1. Introduction to Camera Calibration:

In practical imaging systems, it is often observed that straight lines in the real world may appear curved in images, and objects may seem closer or farther than their actual positions. These effects arise due to imperfections in camera lenses and the imaging process.

Camera calibration is the process of estimating the parameters of a camera that define how three-dimensional (3D) world points are projected onto a two-dimensional (2D) image plane. This process enables accurate interpretation of visual data and reduces discrepancies between real-world geometry and captured images.

An intuitive analogy is that camera calibration functions similarly to an eye examination, where the characteristics of the visual system are measured to ensure accurate perception.

Mathematical Representation

The relationship between a 3D world point and its corresponding 2D image point is given by:

$$x = PX$$

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} p_1 & p_2 & p_3 & p_4 \\ p_5 & p_6 & p_7 & p_8 \\ p_9 & p_{10} & p_{11} & p_{12} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

homogeneous image 3 x 1 Camera matrix 3 x 4 homogeneous world point 4 x 1

where:

- X represents a 3D point in the real world

- x represents the corresponding 2D image point
- P is the camera projection matrix containing all relevant camera parameters

The matrix P defines the transformation from the 3D coordinate system to the 2D image plane and encapsulates both intrinsic and extrinsic parameters of the camera.

Camera Parameters

1- Intrinsic Parameters

Intrinsic parameters describe the internal characteristics of the camera, including:

- Focal Length (f_x, f_y): Determines the scale of projection along the image axes
- Principal Point (c_x, c_y): Represents the optical center of the image
- Skew: Describes the non-orthogonality between image axes
- Aspect Ratio: Defines the relationship between pixel width and height

These parameters collectively define how the camera forms an image from incoming light.

2- Extrinsic Parameters

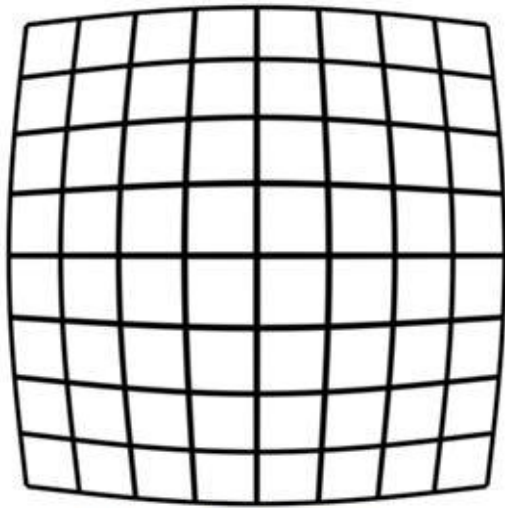
Extrinsic parameters define the spatial relationship between the camera and the world coordinate system:

- Rotation (R): Specifies the orientation of the camera
- Translation (t): Specifies the position of the camera in space

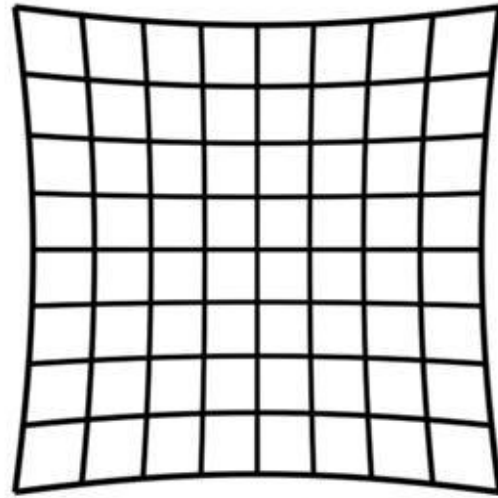
These parameters enable mapping between world coordinates and the camera coordinate system.

Lens Distortion

Real-world lenses introduce distortions that deviate from the ideal pinhole camera model.



Barrel Distortion



Pincushion Distortion

These distortions can be categorized as:

- Radial Distortion: Causes straight lines to appear curved. It is commonly observed as:
 - Barrel distortion (outward bulging)
 - Pincushion distortion (inward pinching)
- Tangential Distortion: Occurs due to misalignment between the lens and the image sensor, leading to asymmetrical image deformation

Accurate modeling and correction of these distortions are essential for precise vision-based measurements.

Importance of Camera Calibration

Camera calibration is critical in applications where geometric accuracy is required. Without proper calibration, measurements derived from images can be significantly inaccurate.

Key benefits include:

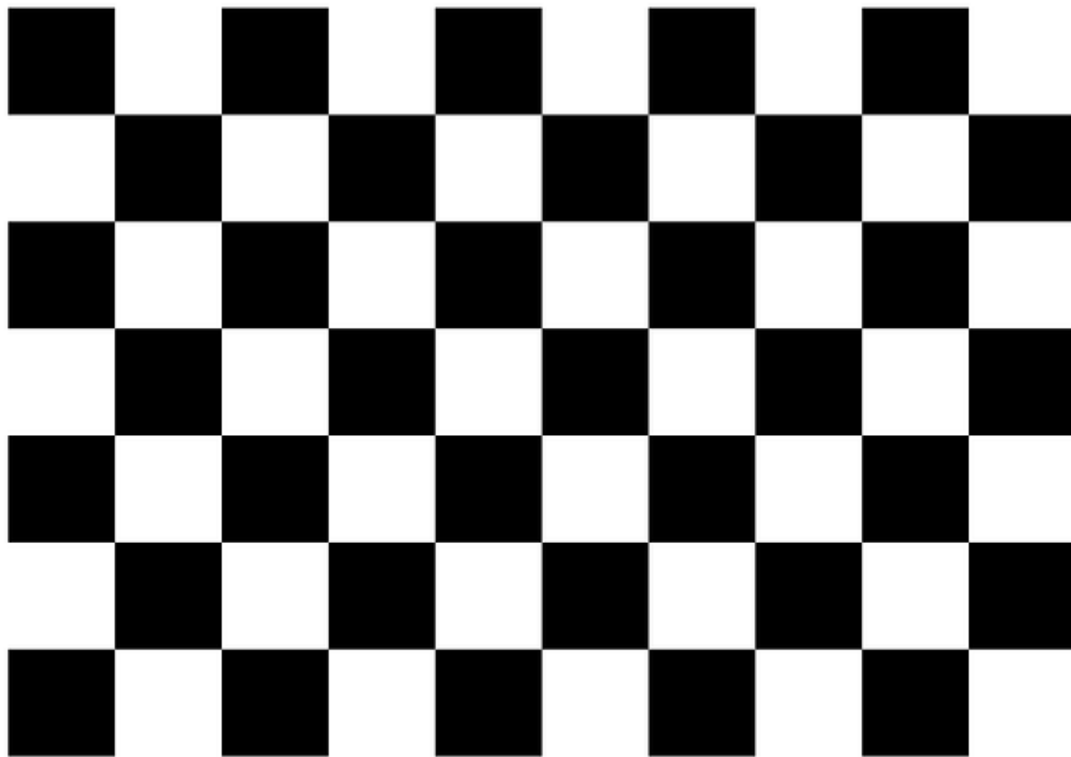
- Distortion Correction: Eliminates lens-induced geometric errors
- Accurate Measurement: Enables reliable distance and dimension estimation

- Enhanced Computer Vision Performance: Improves the reliability of detection and tracking algorithms
- Augmented Reality Alignment: Ensures correct placement of virtual objects in real environments

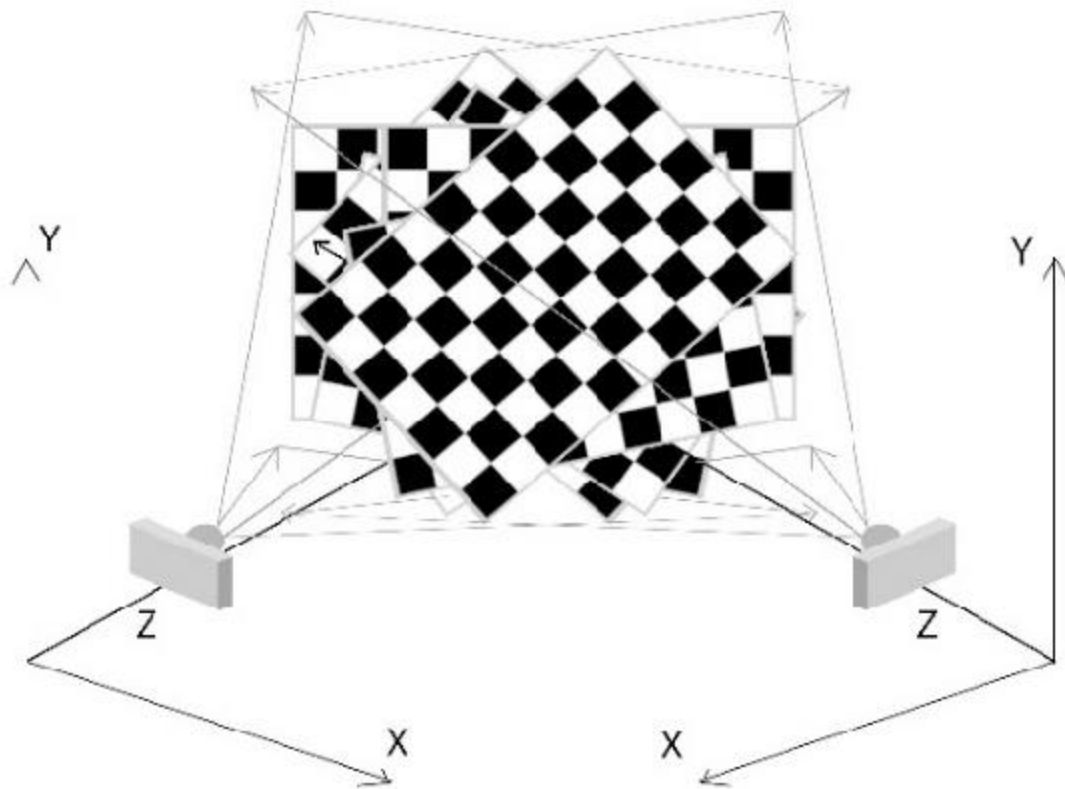
Calibration Procedure Overview

A typical camera calibration process involves the following steps:

1. Preparation of Calibration Target: A checkerboard pattern with known dimensions is commonly used.



2. Image Acquisition: Multiple images are captured from different angles and distances, ensuring full visibility of the pattern.



3. Feature Detection: Corner points of the checkerboard are detected in each image.
4. Parameter Estimation: Using known 3D points and detected 2D points, intrinsic and extrinsic parameters are computed using established algorithms such as Zhang's method.
5. Accuracy Evaluation: Calibration quality is assessed using reprojection error, which measures the difference between observed and projected points.

Practical Considerations

For reliable calibration results:

- Images should be captured from diverse viewpoints
- The calibration pattern must be clearly visible
- Motion blur and poor lighting should be avoided

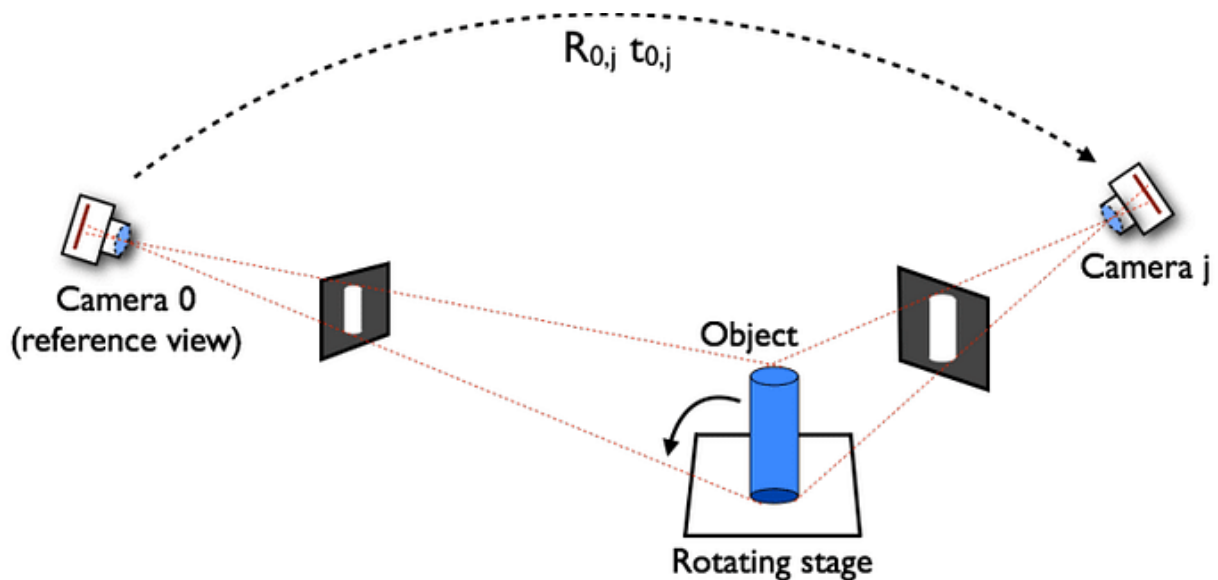
Improper data collection can lead to inaccurate calibration even if the computational process executes successfully.

Applications of Camera Calibration

Camera calibration is a foundational component in multiple domains:

- **3D Reconstruction:**

Camera calibration plays a critical role in 3D reconstruction tasks, particularly in applications such as virtual reality (VR) and augmented reality (AR). By accurately modeling the relationship between multiple 2D images, calibration enables the generation of precise 3D representations of real-world scenes. This is essential in environments such as gaming and simulation, where realistic spatial interactions are required.



- **Robotics:**

In robotic systems, visual input is fundamental for navigation and interaction with the environment. Camera calibration allows robots to estimate object dimensions and distances with improved accuracy. In applications such as autonomous driving, calibrated cameras assist in determining the position of obstacles and surrounding vehicles, thereby supporting safe and reliable operation.

- **Augmented Reality:**

Augmented reality systems rely on accurate alignment between virtual elements and the physical environment. Camera calibration ensures that digital overlays are correctly positioned and scaled relative to real-world objects, resulting in consistent and visually coherent integration.

- **Medical Imaging:**

In medical imaging applications, including endoscopy and radiology, camera calibration is essential for maintaining geometric accuracy. It ensures that captured images accurately represent anatomical structures, which is critical for diagnosis, measurement, and treatment planning.

- **Sports Analysis:**

Camera calibration is widely used in sports analytics to study player motion and game dynamics. By calibrating multiple cameras around a field or arena, it becomes possible to capture accurate spatial and temporal information, enabling detailed performance analysis and improved training strategies.

Functions (OpenCV Reference)

The following OpenCV functions will be used throughout this lab:

- `cv2.findChessboardCorners()`: Detects approximate locations of checkerboard corner points in an image.
- `cv2.cornerSubPix()`: Refines detected corner locations to sub-pixel accuracy for better precision.
- `cv2.calibrateCamera()`: Computes camera intrinsic parameters, distortion coefficients, and extrinsic parameters.
- `cv2.projectPoints()`: Projects 3D object points back onto the image plane; used for error calculation.
- `cv2.undistort()`: Applies distortion correction to an image using calibration parameters.
- `cv2.initUndistortRectifyMap()` and `cv2.remap()`: Used for efficient undistortion, especially in real-time applications (e.g., video).

Lab Tasks

Task 1: Load a checkerboard image and convert it to grayscale. Detect the checkerboard corners using `cv2.findChessboardCorners()` and refine their positions using `cv2.cornerSubPix()`. Visualize the detected corners by overlaying them on the image to verify correct detection.

Task 2: Assume a known square size (e.g., 30 mm) and define the corresponding 3D object points of the checkerboard. Use the detected image points to perform camera calibration with `cv2.calibrateCamera()`. Compute and report the camera matrix (K) and distortion coefficients (k_1, k_2, p_1, p_2, k_3), and observe the obtained parameter values.

Task 3: Capture at least two checkerboard images from different viewpoints. Detect and refine corners for all images as performed earlier. Use these multiple observations to perform calibration

and compute an improved camera matrix. Apply `cv2.getOptimalNewCameraMatrix()` and use the results to undistort an image. Compare the calibrated output with the original image.

Task 4: Load a test image along with the previously computed camera matrix and distortion coefficients. Apply `cv2.undistort()` to remove lens distortion. Compare the original and corrected images side by side, focusing on geometric improvements such as straight lines.

Task 5: After calibration, use `cv2.projectPoints()` to project the 3D object points back onto the image plane. Compare these projected points with the originally detected corner points. Compute the mean reprojection error and use it as a measure of calibration accuracy (lower values indicate better results, typically below 0.5 pixels).